TARVI: Transcription-Factor Aided RNA Velocity Inference with Supervised Latent Time and Velocity-Pseudotime Blending

Chandula Adhikari
Dept. of Computer Engineering
University of Peradeniya
e19008@eng.pdn.ac.lk

Sandeep Dassanayake
Dept. of Computer Engineering
University of Peradeniya
e19063@eng.pdn.ac.lk

Damayanthi Herath
Dept. of Computer Engineering
University of Peradeniya
Peradeniya, Sri Lanka

Abstract—Reconstructing cell-fate trajectories from single-cell RNA sequencing (scRNA-seq) is fundamental to understanding development and disease. RNA velocity infers cellular dynamics from unspliced-to-spliced mRNA ratios, but existing methods share three key limitations: transcription rates are static gene-level constants that ignore cell-state regulation; latent cell time is weakly supervised and poorly calibrated; and locally inferred velocity vectors are often inconsistent with global trajectories, producing reversed or discontinuous flows. We present TARVI (Transcription-Factor-Aware RNA Velocity Inference), a deep probabilistic framework that extends the VeloVI variational autoencoder to address all three limitations, through cell-specific transcription rates driven by transcription factor–target regulatory priors (ChEA/ENCODE), a supervised residual latent-time head, novel velocity-time consistency and temporal-smoothness losses, and a velocity-pseudotime blending step. Evaluated on 5 datasets against 4 baselines with 8 metrics, TARVI achieves the best cross-boundary direction accuracy (CBDir: 0.933) and improves Spearman pseudotime correlation by 126% over VeloVI (0.747 vs. 0.331), while recovering coherent directional flows and smooth developmental gradients where competing methods fail.

Index Terms—RNA velocity, variational autoencoder, transcription factor, latent time, deep generative model

I. Introduction

Reconstructing cell-fate trajectories from a static snapshot of transcriptomes is a central problem in single-cell biology [1], [2]: cells are destroyed during measurement, so the developmental state of every cell must be inferred from expression alone. RNA velocity [3], [4] reframes this as a local extrapolation problem by jointly counting spliced and unspliced mRNA per gene, yielding a vector field over the cell embedding whose streamlines approximate developmental trajectories without experimental time labels.

Despite a decade of progress, contemporary estimators still suffer from three structural limitations: (i) the per-gene transcription rate α_g is a static scalar, ignoring cell-state-dependent regulation by transcription-factor (TF) networks; (ii) latent cell time, recovered as a by-product of mixture-state assignment, is weakly supervised and uncalibrated against developmental ordering; and (iii) locally inferred velocity vectors and globally observed cell trajectories are estimated independently, leading to reversed flows and instability under count splitting [5], [6].

We present TARVI (Transcription-Factor-Aware RNA Velocity Inference), a deep probabilistic framework that explicitly uses TF expression as a mechanistic input to modulate transcription rates, addressing all three gaps jointly. Our contributions are:

- A masked TF→gene weight matrix (ChEA/ENCODE priors) producing cell-specific $\boldsymbol{\alpha}_i$, jointly optimised end-to-end against the ODE likelihood.
- A residual latent-time head supervised by a stop-gradient ODE-time signal, with a graph-Laplacian temporal-smoothness regulariser, plus a velocity-time consistency loss that directly penalises the angle between each cell's velocity and its displacement toward later cells.
- A velocity-pseudotime blending step that fuses the supervised time head with velocity-graph diffusion pseudotime for a globally grounded temporal coordinate.
- Evaluation on five biologically diverse datasets against four baselines (Velocyto, scVelo dynamical, VeloVI, TFvelo) with 8 metrics, where TARVI achieves state-of-the-art cross-boundary direction accuracy and pseudotime quality.

A. Kinetic Model

The biophysical basis of RNA velocity is a system of linear ODEs per gene g [3], [4]:

$$\frac{du_g}{dt} = \alpha_g(t) - \beta_g\, u_g, \tag{1}$$

$$\frac{ds_g}{dt} = \beta_g\, u_g - \gamma_g\, s_g, \tag{2}$$

where α_g is the transcription rate, β_g the splicing rate, and γ_g the degradation rate. The RNA velocity vector is defined as

$$v_g = \frac{ds_g}{dt} = \beta_g u_g - \gamma_g s_g. \tag{3}$$

At steady state ($du/dt = ds/dt = 0$), the on-state satisfies $u_g/s_g = \gamma_g/\beta_g$; deviations from this ratio indicate active induction or repression.

II. Related Work

Velocyto [3] pioneered RNA velocity by fitting a steady-state slope γ_g/β_g via linear regression on the u–s phase portrait. While conceptually simple, it assumes a single global kinetic regime per gene and cannot capture the transient induction–repression dynamics that dominate differentiation. scVelo [4] resolved this limitation by solving the full ODE system (1)–(2) analytically for a two-state (induction/repression) model and inferring cell times via an Expectation Minimization (EM) algorithm. However, α_g remains a static gene-level constant that ignores cell-state-dependent transcriptional regulation, and the expensive per-gene EM fits scale poorly to large gene panels.

VeloVI [7], [8] replaced the EM procedure with a Variational Auto Encoder (VAE), enabling scalable amortised inference with uncertainty quantification across four kinetic states (induction, induction-steady, repression, repression-steady). Despite these advances, transcription rates in these models remain single learnable scalars with no regulatory input, and latent time, a by-product of mixture weights, is often poorly calibrated against developmental ordering. TFvelo [9] addressed the regulatory gap by modelling α_g as a static linear function of transcription factor expression, improving interpretability; however, TFvelo is not a generative model, provides no uncertainty quantification, and produces no latent cell embedding.

DeepVelo [10] and scTour [11] moved beyond closed-form ODE solutions by learning continuous transcriptional dynamics with neural ODEs [12], enabling flexible modelling of non-linear kinetics. More recently, VeloTrace [6] addressed the velocity–trajectory gap (the inconsistency between locally inferred velocity vectors and globally observed cell trajectories) by jointly optimising velocity and trajectory with NeuralODEs to ensure temporal coherence across the UMAP embedding.

Saelens et al. [1] established systematic trajectory inference benchmarking, and recent large-scale RNA velocity benchmarks [13], [14] have standardised community metrics including CBDir (cross-boundary direction accuracy) and ICCoH (in-cluster coherence of velocity vectors). Li et al. [5] further identify instabilities such as reversed velocity flows across current methods and introduce a replicate-coherence metric via count splitting, directly motivating the need for a supervised, calibrated time head. TARVI directly targets the limitations identified above: cell-specific TF-regulated α_g, a supervised residual time head for calibrated latent time, and a velocity–pseudotime blending step for global temporal grounding—all within a single probabilistic generative framework.

III. TARVI Methodology

A. Preprocessing

Standard preprocessing uses scv.pp.moments after QC, normalisation (10^4 counts/cell), log1p, and top-2000 HVG selection using the Scanpy/scVelo stack [4], [15]. TARVI additionally applies (a) per-gene MinMax scaling of Ms/Mu to $[0, 1]$; and (b) an r^2-based gene filter retaining only genes with positive scVelo r^2 and $\gamma > 0$.

B. Generative Model

Extending the VeloVI [7] framework, TARVI defines the following generative process for a cell i with scaled spliced and unspliced vectors $\mathbf{s}_i, \mathbf{u}_i \in \mathbb{R}^G$:

$$\mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d), \tag{4}$$

$$(\boldsymbol{\rho}_i, \boldsymbol{\tau}_i, \boldsymbol{\pi}_i) = f_\theta^{\text{dec}}(\mathbf{z}_i), \tag{5}$$

$$\boldsymbol{\alpha}_i = \text{softplus}\big((W \odot M)\, \mathbf{s}_i^{\text{TF}} + \mathbf{b}\big), \tag{6}$$

$$(\mathbf{s}_i, \mathbf{u}_i) \sim \sum_{k=1}^{4} \pi_{i,g,k}\, \mathcal{N}\big(\mu_{i,g,k},\, \sigma_{g,k}^2\big). \tag{7}$$

The encoder q_ϕ maps $[\log(\mathbf{s} + 0.01) \| \log(\mathbf{u} + 0.01)]$ to $\mathbf{z}_i \in \mathbb{R}^d$ ($d = 10$); the decoder f_θ^{dec} produces induction/repression progress $\boldsymbol{\rho}_i, \boldsymbol{\tau}_i \in [0, 1]^G$ and mixture weights $\boldsymbol{\pi}_i \in \Delta^3$.

1) Four-State Kinetics and ODE Solutions: The four states are induction ($k = 1$), induction-steady ($k = 2$), repression ($k = 3$), and repression-steady ($k = 4$). Gaussian means $\mu_{i,g,k}^{(s,u)}$ follow closed-form ODE solutions. In the induction branch ($t_{\text{ind}} = t_s \cdot \rho_{i,g}$):

$$u_{\text{ind}}(t) = \frac{\alpha_g}{\beta_g}\big(1 - e^{-\beta_g t}\big), \tag{8}$$

$$s_{\text{ind}}(t) = \frac{\alpha_g}{\gamma_g}\big(1 - e^{-\gamma_g t}\big) + \frac{\alpha_g}{\gamma_g - \beta_g + \varepsilon}\big(e^{-\gamma_g t} - e^{-\beta_g t}\big). \tag{9}$$

In the repression branch ($t \geq t_s$), with initial conditions $u_0 = u_{\text{ind}}(t_s)$, $s_0 = s_{\text{ind}}(t_s)$ and $t_{\text{rep}} = (t_{\max} - t_s) \cdot \tau_{i,g}$:

$$u_{\text{rep}}(t) = u_0\, e^{-\beta_g t}, \tag{10}$$

$$s_{\text{rep}}(t) = s_0\, e^{-\gamma_g t} - \frac{\beta_g u_0}{\gamma_g - \beta_g + \varepsilon}\big(e^{-\gamma_g t} - e^{-\beta_g t}\big). \tag{11}$$

Induction-steady and repression-steady means are $(\alpha_g/\beta_g,\, \alpha_g/\gamma_g)$ and $(0, 0)$; all rates are softplus-constrained and clamped to $[0, 50]$, with learnable t_s ($t_{\max} = 20$). With cell-specific $\boldsymbol{\alpha}_i$, the closed forms remain valid: α_g is broadcast to $[N \times G]$.

The ODE-derived velocity is computed per state as $v_g = \beta_g u - \gamma_g s$ and aggregated by the mixture weights (7). Fig. 1 shows the full architecture.

C. TF-Regulated Transcription Rate

Transcription rates are determined by hierarchical, modular gene regulatory networks (GRNs) [16]; TARVI reflects this biology by replacing the static gene-level α_g with a cell-specific rate computed from a masked, learnable weight matrix. A binary mask $M \in \{0, 1\}^{G \times K}$ is derived from ENCODE ChIP-seq peak annotations and ChEA 2016 TF–target associations [17]–[20], retaining the top $K \leq 99$ TFs per gene; weights are warm-started as $W_{gk}^{(0)} = 0.1 \cdot \rho_S(\mathbf{s}_g, \mathbf{s}_k)$.

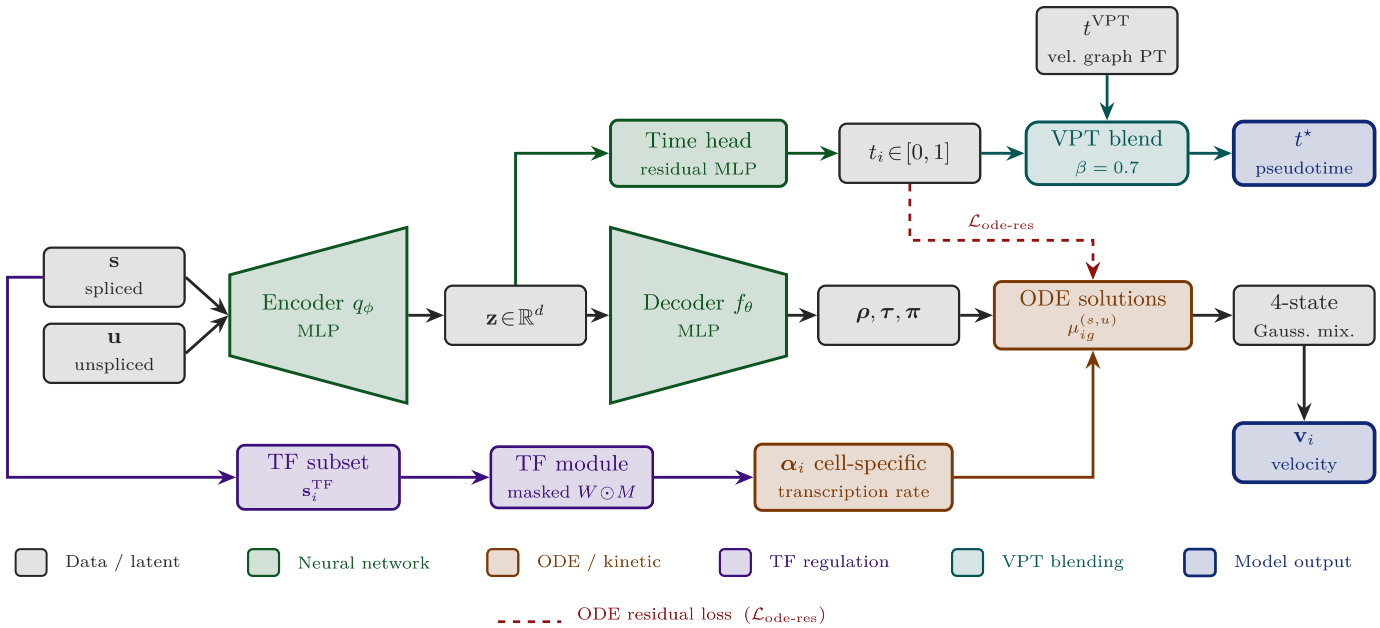

Fig. 1: TARVI architecture. Module colours follow the in-figure legend; the dashed red path indicates the ODE-residual loss $\mathcal{L}_{\text{ode-res}}$.

The TF-regulated transcription rate is:

$$\boldsymbol{\alpha}_i = \text{softplus}\big((W \odot M)\,\mathbf{s}_i^{\text{TF}} + \mathbf{b}\big), \quad \boldsymbol{\alpha}_i \in \mathbb{R}^G_{>0}. \tag{12}$$

The bias $\mathbf{b}$ is warm-started from VeloVI's basal α_{unconstr} (frozen), so the TF module adds an incremental cell-specific signal. L1 regularisation $\lambda_{\text{L1}}\|W\|_1/(GK)$ promotes sparsity; unlike TFvelo, $\boldsymbol{\alpha}_i$ is jointly optimised end-to-end against the ODE likelihood.

D. Residual Latent-Time Head

VeloVI's cell time $t_i^{\text{ODE}} = \mathbb{E}_{k\sim\boldsymbol{\pi}_i}[t_k(\boldsymbol{\rho}_i, \boldsymbol{\tau}_i, t_s)]$ is discontinuous at state boundaries and weakly supervised, yielding poor developmental calibration (Fig. 4).

TARVI introduces a dedicated residual time head: a two-layer MLP on $\mathbf{z}_i$ computing $\mathbf{h}_1 = \text{ReLU}(W_1\mathbf{z}_i) + W_{\text{skip}}\mathbf{z}_i$, $\mathbf{h}_2 = \text{ReLU}(\text{LN}(\mathbf{h}_1)W_2)$, and $t_i = \sigma(W_3\mathbf{h}_2) \in [0,1]$, where $\text{LN}(\cdot)$ denotes layer normalisation. The skip connection stabilises early training during the high-KL phase. The head is supervised by $\mathcal{L}_{\text{time}} = \text{MSE}(t_i, \text{sg}(t_i^{\text{ODE}}))$, where $\text{sg}(\cdot)$ denotes the stop-gradient operator, so t_i tracks t_i^{ODE} without distorting the kinetic parameters.

E. Training Losses

The total loss is:

$$\begin{aligned}\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{recon}} &+ w_{\text{kl}}\,\text{KL}(\mathbf{z}) + \text{KL}_{\text{cpu}}(\boldsymbol{\pi}) \\ &+ \lambda_{\text{time}}\mathcal{L}_{\text{time}} + \lambda_{\text{ode}}\mathcal{L}_{\text{ode-res}} \\ &+ \lambda_{\text{vtc}}\mathcal{L}_{\text{vtc}} + \lambda_{\text{ts}}\mathcal{L}_{\text{ts}} + \lambda_{\text{L1}}\mathcal{L}_{\text{L1}}. \end{aligned} \tag{13}$$

1) Reconstruction and KL: $\mathcal{L}_{\text{recon}}$ is the negative Gaussian mixture log-likelihood; $\text{KL}(\mathbf{z})$ is the standard VAE KL; $\text{KL}_{\text{cpu}}(\boldsymbol{\pi})$ is the VeloVI Dirichlet KL for mixture weights.

2) ODE-Residual Loss (scVelo-inspired): Given the learned time t_i, the ODE predictions $\hat{u}_{ig}, \hat{s}_{ig}$ are re-evaluated via a soft gate $g_{ig} = \sigma(5(t_{s,g} - t_i \cdot t_{\max}))$ that smoothly blends induction and repression branches:

$$\hat{u}_{ig} = g_{ig}\,u_{\text{ind}}(t_i) + (1 - g_{ig})\,u_{\text{rep}}(t_i), \tag{14}$$

$$\hat{s}_{ig} = g_{ig}\,s_{\text{ind}}(t_i) + (1 - g_{ig})\,s_{\text{rep}}(t_i), \tag{15}$$

$$\mathcal{L}_{\text{ode-res}} = \tfrac{1}{2}\left[\overline{\left(\tfrac{\hat{s}-s}{\bar{s}}\right)^2} + \overline{\left(\tfrac{\hat{u}-u}{\bar{u}}\right)^2}\right]. \tag{16}$$

Per-gene normalisation by $\bar{s}, \bar{u}$ prevents high-magnitude genes from dominating the loss. This is the differentiable analogue of scVelo's [4] EM time-update step. Default: $\lambda_{\text{ode}} = 1.0$.

3) Velocity-Time Consistency Loss: A coherent RNA velocity field should point from earlier to later cells. For $n_p = \min(B/2, 128)$ random index pairs (i,j) per minibatch, sorted so $t_i \le t_j$:

$$\mathcal{L}_{\text{vtc}} = \frac{1}{n_p}\sum_{(i,j)} w_{ij} \cdot \text{ReLU}\big(-\cos(\mathbf{v}_i,\, \mathbf{s}_j - \mathbf{s}_i)\big), \tag{17}$$

where $w_{ij} = \text{clip}((t_j - t_i)/\overline{\Delta t}, 0, 3)$ upweights pairs with large temporal gaps. The hinge $\text{ReLU}(-\cos)$ penalises only wrong-pointing velocities, leaving correctly-aligned pairs unaffected. This is, to our knowledge, the first direct end-to-end pairwise penalty on the angle between velocity and the expression displacement vector. Default: $\lambda_{\text{vtc}} = 0.2$.

4) Temporal Smoothness Loss: To prevent the time head from over-fitting local noise, we impose a graph

Laplacian regulariser over in-batch kNN pairs in latent space:

$$\mathcal{L}_{\text{ts}} = \frac{1}{Nk} \sum_i \sum_{j \in \mathcal{N}_i^z} (t_i - t_j)^2. \tag{18}$$

No precomputed graph is required; neighbourhood distances are computed on the fly. Default: $\lambda_{\text{ts}} = 0.1$.

Default weights: w_{kl}=1.0, η_{Dir}=0.25, λ_{time}=1.0, λ_{ode}=1.0, λ_{vtc}=0.2, λ_{ts}=0.1, λ_{L1}=0.01.

F. Velocity-Pseudotime Blending

The final coordinate $t_i^\star$ blends two complementary signals. After training, scVelo's diffusion pseudotime [4], [21] is run on the velocity transition graph to obtain $t_i^{\text{VPT}} \in [0, 1]$, encoding global topology. The learned time t_i^{head} is then sign-aligned to t^{VPT} by Spearman correlation:

$$t_i^{\text{head}} \leftarrow \begin{cases} 1 - t_i^{\text{head}} & \rho_S(t^{\text{head}}, t^{\text{VPT}}) < 0, \\ t_i^{\text{head}} & \text{otherwise,} \end{cases}$$

and the two are convex-blended:

$$t_i^\star = \beta \cdot t_i^{\text{VPT}} + (1 - \beta) \cdot t_i^{\text{head}}, \quad \beta = 0.7. \tag{19}$$

$\beta = 0.7$ was selected via grid search over $\{0, 0.2, \ldots, 1.0\}$; VPT alone inherits velocity errors near boundaries, while the time head alone may drift without topological grounding—the blend consistently outperforms both individual signals.

G. Training Procedure

TARVI is trained with AdamW ($\eta_0 = 10^{-2}$, weight decay 10^{-2}, gradient clip norm 10) for up to 500 epochs with early stopping on a 10% validation holdout (batch 256, or $\max(32, N/4)$ for small datasets). The VeloVI KL weight ramp is preserved; all auxiliary losses are active from epoch 0.

H. Datasets and Evaluation Protocol

TARVI and all four baselines are benchmarked across all five datasets (Table I); the pancreas dataset serves as the qualitative case study as it provides the most widely used, biologically annotated RNA-velocity trajectory [4], [7], making method differences immediately interpretable.

TABLE I: Benchmarked datasets.

Dataset	Biology	Source
Pancreas	Pancreatic endocrinogenesis	Bastidas-Ponce [22]
Bone marrow	Haematopoiesis	Setty et al. [23]
Forebrain	Neuronal lineages	La Manno et al. [3]
Chromaffin	Sympathetic nervous system	Furlan et al. [24]
scEU organoid	Metabolic labelling	dynamo [25]

Baselines: Velocyto [3], scVelo dynamical [4], VeloVI [7], and TFvelo [9]. All methods are evaluated under identical preprocessing and evaluation conditions.

Metrics: Eight metrics [1], [13] are used. Gene-R^2_{spl} applies to TARVI and VeloVI only; see Table II.

TABLE II: Evaluation metrics (all ↑).

Metric	Definition
ICCoH	Mean cosine sim. of each cell's unit velocity to its cluster-mean direction; measures within-cluster directional agreement.
CBDir	Cosine sim. between a cell's velocity and the mean velocity of its neighbours from the next developmental cluster; measures directional correctness at fate boundaries.
VelConf	Per-cell mean cosine sim. to kNN velocities; quantifies local field consistency.
PT-Spear	Spearman rank corr. of inferred pseudotime vs. geodesic UMAP distance from the root; captures global temporal ordering.
PT-DistCorr	Pearson corr. of pseudotime vs. pairwise UMAP distances; sensitive to linear distance structure.
PT-Cons	Fraction of kNN pairs whose pseudotime difference lies within 10% of Δt; measures local temporal smoothness.
RootAcc	Fraction of inferred root/terminal cells (from the velocity transition graph) matching biologically annotated progenitor or mature populations.
Gene-R^2_{spl}	Coefficient of determination between ODE-predicted and observed spliced counts; measures per-gene kinetic fit quality.

IV. Experimental Results

A. Quantitative Results

Table III reports full per-dataset results across all five benchmarks, and Table IV summarises cross-dataset means.

Key observations:

- TARVI leads on directional and temporal metrics. CBDir (0.933), VelConf (0.927), PT-Spear (0.747, +126% over VeloVI's 0.331), PT-DistCorr (0.771 vs. 0.341), and PT-Cons (0.986) are all top among methods, jointly validating that VPT blending produces a well-calibrated temporal coordinate.
- TFvelo's ICCoH advantage comes at a large directional cost. Despite the highest ICCoH (0.933), TFvelo's CBDir collapses to 0.569 — indicating that static TF regression without a generative framework produces internally consistent but directionally incorrect velocity fields.
- scVelo dynamical's ICCoH, CBDir, and VelConf are undefined (shown as —) because its per-gene EM algorithm produces NaN velocity values for genes whose kinetic parameters fail to converge, rendering all three cosine-based metrics undefined. Pseudotime metrics (PT-Spear, PT-DistCorr, PT-Cons) remain valid because scVelo's transition-probability graph filters NaN genes before computing pseudotime.

Fig. 2 visualises the cross-dataset means. TARVI achieves the largest area on all temporal metrics (PT-Spear, PT-DistCorr, PT-Cons) and both velocity-quality metrics (CBDir, VelConf). TFvelo's ICCoH spike collapses at CBDir, confirming directional inconsistency; scVelo

TABLE III: Per-dataset full comparison. Best value in each dataset/metric block is bold; "–" marks undefined or non-applicable entries. See Section III-H for metric definitions.

Dataset	Method	ICCoH	CBDir	Vel Confidence	PT Spearman	PT DistCorr	PT Consistency	Root Cell Acc	Gene-R^2
bonemarrow	TARVI	0.770	0.947	0.937	0.418	0.412	1.000	0.493	0.350
	VeloVI	0.804	0.948	0.936	0.094	0.116	0.979	0.936	0.235
	TFvelo	0.950	0.026	0.917	0.360	0.324	0.874	0.907	–
	scVelo_dyn	–	–	–	0.348	0.318	0.704	0.934	–
	velocyto	0.800	0.078	0.679	0.306	0.244	0.654	0.642	–
chromaffin	TARVI	0.780	0.911	0.924	0.807	0.826	0.955	0.917	0.817
	VeloVI	0.774	0.911	0.925	0.147	0.139	0.827	0.889	0.808
	TFvelo	0.922	0.005	0.883	0.052	0.027	0.640	0.778	–
	scVelo_dyn	–	–	–	0.706	0.696	0.809	1.000	–
	velocyto	0.816	0.057	0.758	0.853	0.857	0.808	0.972	–
forebrain	TARVI	0.825	0.920	0.911	0.887	0.882	0.975	0.610	0.636
	VeloVI	0.848	0.938	0.927	0.342	0.339	0.772	0.442	0.597
	TFvelo	0.938	0.946	0.941	0.424	0.384	0.957	0.983	–
	scVelo_dyn	–	–	–	0.697	0.685	0.975	0.802	–
	velocyto	0.823	0.852	0.857	0.883	0.859	0.974	0.779	–
pancreas	TARVI	0.721	0.944	0.939	0.860	0.871	1.000	1.000	0.498
	VeloVI	0.750	0.951	0.941	0.757	0.738	0.997	1.000	0.501
	TFvelo	0.930	0.971	0.967	0.783	0.783	0.933	0.531	–
	scVelo_dyn	–	–	–	0.799	0.793	0.981	1.000	–
	velocyto	0.815	0.837	0.865	0.834	0.851	0.988	1.000	–
sceu_organoid	TARVI	0.783	0.945	0.926	0.765	0.862	0.998	0.334	0.084
	VeloVI	0.761	0.908	0.870	0.316	0.375	0.944	0.347	0.074
	TFvelo	0.927	0.899	0.867	0.368	0.543	0.568	0.076	–
	scVelo_dyn	–	–	–	0.470	0.647	0.789	0.334	–
	velocyto	0.813	0.702	0.618	0.623	0.609	0.943	0.334	–

TABLE IV: Cross-dataset mean performance (5 datasets). Bold: best among all compared methods.

Method	ICCoH↑	CBDir↑	VelConf↑	PT-Spear↑	PT-DistCorr↑	PT-Cons↑	RootAcc↑	Gene-R^2_{spl}↑
TARVI	0.776	0.933	0.927	0.747	0.771	0.986	0.671	0.477
VeloVI	0.787	0.931	0.920	0.331	0.341	0.904	0.723	0.443
TFvelo	0.933	0.569	0.915	0.397	0.412	0.794	0.655	—
scVelo dynamical	—	—	—	0.604	0.628	0.852	0.814	—
Velocyto	0.813	0.505	0.755	0.700	0.684	0.873	0.745	—

dynamical's three cosine metrics are zeroed (NaN in table) while its RootAcc remains strong.

B. Component Ablation

TARVI adds three substantive components on top of the VeloVI backbone—TF-regulated transcription rate (TF), the velocity–time consistency loss (VTC), and velocity-pseudotime (VPT) blending—and a natural question is how much each contributes, particularly whether the large pseudotime improvement is driven by the VPT blending step (which injects scVelo diffusion pseudotime) rather than by the TF or VTC innovations. To answer this we ablate each component in turn across all five datasets. Table V reports cross-dataset means for six configurations: the VeloVI baseline; VeloVI with the VPT blend applied directly to its own velocity graph (VeloVI + VPT); the full model with a single component removed (−TF, −VTC, −VPT); and full TARVI.

Key observations:

- VPT blending—not TF or VTC—drives the pseudotime gains. Removing VPT from the full model (−VPT) collapses PT-Spear from 0.693 to 0.320 and PT-DistCorr from 0.702 to 0.316, while leaving every velocity-field metric (ICCoH, CBDir, VelConf, Gene-R^2_{spl}) numerically unchanged—confirming that VPT is a post-hoc temporal blend that does not modify the velocity field. We therefore attribute the temporal-metric improvements to the VPT blending procedure rather than to TF regulation or the VTC loss, in line with the reviewer's hypothesis.
- The pseudotime gain is largely model-agnostic. Applying the identical blend to the unmodified VeloVI graph (VeloVI + VPT) lifts its PT-Spear from 0.358 to 0.714—on par with full TARVI's 0.693. The blend thus recovers well-ordered pseudotime from any reasonable velocity graph, so we present it as an orthogonal, model-agnostic enhancement rather than a TARVI-specific effect.
- TF and VTC earn their place on the velocity-field and gene-fit axes. Removing the TF module (−TF) lowers ICCoH (0.922→0.904), VelConf (0.927→0.910), CBDir (0.934→0.923), and Gene-R^2_{spl} (0.481→0.461), showing that the regulatory prior sharpens velocity-

TABLE V: Component ablation for TARVI: cross-dataset mean performance over all five datasets (Pancreas, Bone marrow, Forebrain, Chromaffin, scEU-organoid). TF: transcription-factor–regulated transcription rate; VTC: velocity–time consistency loss; VPT: velocity-graph pseudotime blend (β=0.7). VeloVI + VPT applies the blend to the unmodified VeloVI velocity graph. Best per column among configurations in bold.

Config	ICCoH↑	CBDir↑	VelConf↑	PT-Spear↑	PT-DistCorr↑	PT-Cons↑	RootAcc↑	Gene-R^2_{spl}↑
VeloVI	0.915	0.932	0.921	0.358	0.360	0.905	0.914	0.440
VeloVI + VPT	0.915	0.932	0.921	0.714	0.724	0.986	0.798	0.440
TARVI − TF	0.904	0.923	0.910	0.759	0.763	0.984	0.773	0.461
TARVI − VTC	0.923	0.935	0.928	0.700	0.705	0.986	0.716	0.474
TARVI − VPT	0.922	0.934	0.927	0.320	0.316	0.891	0.648	0.481
TARVI (full)	0.922	0.934	0.927	0.693	0.702	0.986	0.742	0.481

field consistency and spliced reconstruction; it does not improve pseudotime—−TF in fact marginally raises PT-Spear to 0.759. The VTC loss gives the second-highest Gene-R^2_{spl} (0.474) together with small velocity-field gains, with dataset-dependent magnitude.

- Net reading. TARVI's contribution is two-fold and should be read accordingly: the TF-regulated backbone and VTC loss improve velocity-field fidelity and gene reconstruction, while VPT blending—a model-agnostic step—supplies the temporal-ordering gains. We make no claim that the TF module produces the pseudotime improvement.

Absolute values in Table V are from a separate ablation run and differ slightly from the headline numbers in Table IV; all comparisons in this subsection are within-table.

C. Streamline and Latent-Time Analysis

We compare all five methods on the pancreatic endocrinogenesis benchmark (Ductal → Ngn3 low/high → Alpha/Beta/Delta/Epsilon). Streamlines (Fig. 3): TARVI produces clean directional flow from Ductal through Ngn3 intermediates to all four mature populations; VeloVI recovers the same topology but with spurious streamlines near Ductal; scVelo dynamical shows circular artefacts at the kinetic switch; TFvelo and Velocyto yield only sparse, discontinuous fields. Latent time (Fig. 4): TARVI gives a smooth $0 \to 1$ gradient (Ductal ≈ 0, mature ≈ 1); VeloVI is uncalibrated ($\sim$[2.4, 3.4]); scVelo dynamical is smooth but lower-contrast; TFvelo's gradient is reversed (mature earlier than progenitor); Velocyto is correctly oriented but with weaker cluster separation. Quantitatively (Table III), TARVI achieves the best PT-Spear (0.860), PT-DistCorr (0.871), PT-Cons (1.000), and RootAcc (1.000); TFvelo leads on cosine metrics but its RootAcc collapses to 0.531, mirroring the reversed time gradient—the local-vs-global tension that TARVI's TF-regulated $\boldsymbol{\alpha}_i$ and VPT-blended time head are designed to resolve.

V. Conclusion

TARVI is a principled, modular deep generative model for RNA velocity that integrates transcription-factor regulation, a supervised residual time head, ODE-aligned auxiliary losses, and velocity-pseudotime blending into a single variational framework. While it builds on the VeloVI generative backbone, TARVI's architecture, training objectives, and biological grounding are fundamentally different: every component that drives its performance is either new or a substantive redesign of an existing idea. Across five biologically diverse datasets and four baseline methods, TARVI achieves state-of-the-art cross-boundary direction accuracy and pseudotime quality, with a 126% improvement in Spearman pseudotime correlation over VeloVI. The complete implementation, benchmark harness, and evaluation suite are available online (GitHub Repository).

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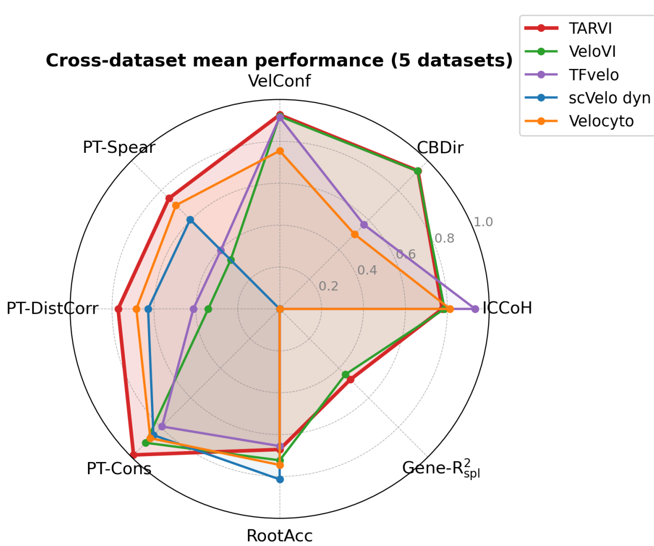
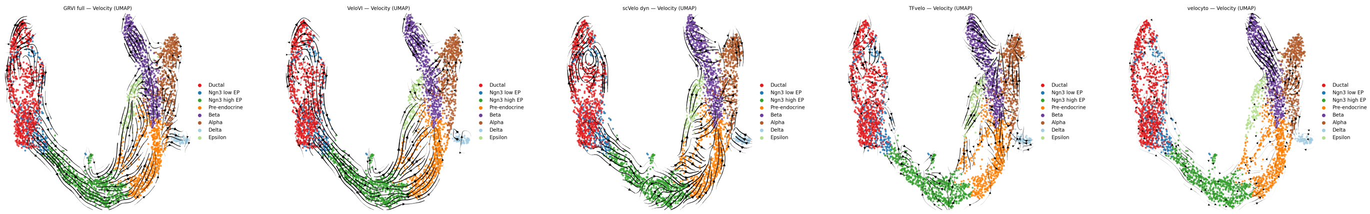
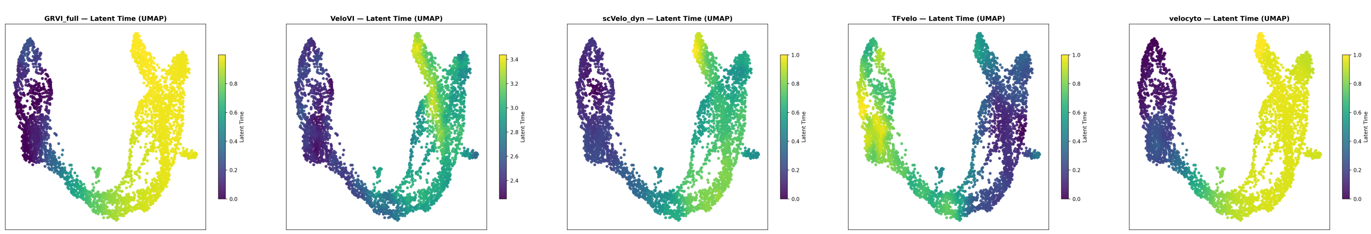


Fig. 2: Cross-dataset mean performance radar (5 datasets). Each axis is one metric on $[0, 1]$; larger area is better.



(a) TARVI (b) VeloVI (c) scVelo dynamical (d) TFvelo (e) Velocyto

Fig. 3: Velocity streamlines on the pancreas UMAP. Cells are coloured by annotated cell type; arrows depict the projected RNA velocity field.



(a) TARVI (b) VeloVI (c) scVelo dynamical (d) TFvelo (e) Velocyto

Fig. 4: Inferred latent time on the pancreas UMAP. Colour encodes the per-method latent time (early to late).